Binomial Theorem:

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{r} a^{n-r} b^r + \cdots + b^n$$

General term:

$\binom{n}{r} a^{n-r} b^r, 0 \leq r \leq n$

Eg: Find the term independent of x in the expansion of $\left(2x + \frac{1}{x}\right)^{10}$

$x^{10}, x^8, x^6, x^4, x^2, x^0$

$$\binom{n}{5} a^{n-5} b^5 = \binom{10}{5} (2x)^{10-5} \left(\frac{1}{x}\right)^5 = 252 \times 32x^5 \times \frac{1}{x^5} = 8064$$

Arithmetic Progression:

Eg: 1, 2, 3, 4, 5, …

First term = 1

Common difference = 1

nth term = 1 + 1(n-1) = n

$a =$ first term

$d =$ common difference

$u_n =$ nth term $= a + d(n-1)$

In an arithmetic progression, the difference between successive terms is constant

Sum of first n terms:

$$S_n = \frac{1}{2} n(a+l) = \frac{1}{2} n\{2a + (n-1)d\}$$

Geometric progression:

Eg: 1, 2, 4, 8, 16, …

First term = 1

Common ratio = 2

nth term = 1 x 2^{n-1} = 2^{n-1}

$a =$ first term

$r =$ common ratio

$u_n =$ nth term $= ar^{n-1}$

In a geometric progression, the ratio between successive terms is constant

Sum of first n terms:

$$S_n = \frac{a(1-r^n)}{1-r} \ (r \neq 1)$$

A geometric progression converges when its common ratio is between -1 and 1

If a geometric progression converges, it has a sum to infinity

Eg: 16, 8, 4, 2, 1, 0.5, 0.25, …

This progression will eventually converge as terms get smaller and smaller

$$S_\infty = \frac{a}{1-r} \ (|r| < 1)$$